

Parasitism alters 3 power laws of scaling in a metazoan community: Taylor's law, density-mass allometry, and variance-mass allometry

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SUPPLEMENTARY MATERIALS: Supplementary text, 6 figures, 1 table

Theory: Taylor's law and density-mass allometry predict variance-mass allometry

TL may be written:

$$\text{variance of population density} = a \times (\text{mean population density})^b, \text{ where } a > 0.$$

On logarithmic scales, TL becomes a linear relationship, assuming the variance and mean are positive:

$$\log(\text{variance of population density}) = \log(a) + b \times \log(\text{mean population density}).$$

This relationship holds for the logarithm to any base, and the value of b is independent of the base of the logarithms. We always used base 10. Because b appears as an exponent in the power-law form of TL and as the slope in the linear relationship, b is sometimes referred to as the exponent and sometimes as the slope of TL. The value of b is the same in either usage.

DMA may be written:

$$\text{mean population density} = u \times (\text{mean body mass per individual})^v, \text{ where } u > 0.$$

Substituting DMA into TL predicts VMA:

$$\text{variance of population density} = c \times (\text{mean body mass per individual})^d,$$

where

$$c = au^b > 0, d = bv.$$

We tested whether the coefficients predicted for VMA from the estimated parameters of TL and DMA fell within the confidence intervals of the coefficients obtained by fitting VMA directly to the data.

Detailed methods

The four lakes (Hayes 44°58'59.4"S, 168°48'19.8"E; Tomahawk Lagoon 45°54'06.0"S, 170°33'02.2"E; Tuakitoto 46°13'42.5"S, 169°49'29.2"E; Waihola 46°01'14.1"S, 170°05'05.8"E) were sampled in three seasons (September 2012, January and May 2013). The four lakes were of modest sizes (2.76 km² surface area and 3.1 m mean depth for Lake Hayes; 0.096 km² and 1.1 m for Tomahawk Lagoon; 5.44 km² and 0.95 m for Lake Tuakitoto; 6.35 km² and 1.33 m for Lake Waihola).

In the combined samples of the four lakes, we found 22 taxonomic species of free-living parasitized species, 57 of free-living unparasitized species, and 35 of parasitic species. In this enumeration, we treated each life stage of parasite species (only) as a separate 'species'. For example, we distinguished redia, metacercaria and adult of the trematode *Telogaster opisthorchis* as separate 'species'. For parasites, we measured density as individuals per square meter. This choice of metric made it possible to compare the variability of parasites with that of free-living parasitized and free-living unparasitized species.

The basic unit for which we calculated population density was a sample taken in one place at one time. Samples used different methods for different organisms (fish = seine nets, gillnets, fyke nets depending on the fish species; benthos = surber sampler; demersal species = standardized sweep net sampling; plankton = plankton net tows). All organisms were counted per sample, including metazoan parasites within their hosts, to give a density for each taxon.

In each lake and in each season (4 lakes × 3 seasons = 12 full sets of samples), we collected samples along four localities, i.e. four stretches of the littoral zone. The four sampled localities within each lake were not very distant from each other, although distances between sampling sites varied among lakes due to differences in lake size and shape (365 m minimum distance and 2680 m maximum distance along the shoreline between two sites in Lake Hayes; 100 m and 460 m in Tomahawk Lagoon; 420 m and 1550 m in Lake Tuakitoto; 1350 m and 3000 m in Lake Waihola). For fish, a single estimate of density was obtained for each locality. For benthic and demersal species we obtained 6 samples per locality and for plankton, 4 samples per locality. For each of the 12 lake × season combinations, means and variances were calculated across: 4 samples for the fish species (and for parasites within fish), 24 samples for benthic or demersal species (and for their parasites), and 16 samples for plankton species (which had no metazoan parasites).

Body mass was calculated differently for different types of organisms. Parasites were generally too small to be weighed and varied little in size within each life stage of each taxonomic species, so we averaged measurements of dimensions from a random subsample of individuals. We calculated the volume of one individual of average dimensions based on the most appropriate formula for its shape (e.g. adult nematodes and acanthocephalans = cylinder, adult trematodes = flattened ellipsoid, encysted juvenile trematodes [metacercariae] = spheres). Their volume was converted to mass assuming their density equaled that of water. Most free-living invertebrates

were large enough to be weighed individually (isopods, chironomids, odonates, large trichopteran larvae, adult hemipterans, mollusks, leeches), and we calculated the average mass of an individual. For small free-living invertebrates (amphipods, small trichopteran larvae, oligochaetes, planktonic crustaceans), which varied little intraspecifically, we pooled 5, 10 or 20 conspecific individuals (depending on the species) from random subsamples, weighed them as a group, and from the total mass calculated the average mass of one individual. For fish, each individual was weighed and we calculated average mass of an individual, giving equal weight to each individual. Consequently fish mass for a given species varied across lakes and seasons, while the mass of smaller organisms was treated as constant for each species (or life stage of parasites within a taxonomic species).

Statistics

The Supplementary data file (.txt format, comma-separated variables) gives the number of samples, the minimum sample size (minimum number of organisms), the maximum sample size (maximum number of organisms), and the total sample size (total number of organisms) for each estimate of the sample mean and the sample variance. The data column "Life stage" uses these abbreviations: Ad = adult, L = larva, C = cystacanth; Mc = metacercaria; Sp = sporocyst; Rd = redia.

In summary, of the 733 estimates of the mean and the variance of population density specific to species, life stage, lake, and season, 158 estimates were based on 4 samples each, 85 estimates were based on 16 samples each and 490 estimates were based on 24 samples each. The total number of samples was 13,752, giving an average of approximately 18.76 samples per estimate of mean and variance.

The total number of organisms was 1,037,058, of which 518,295 were free-living unparasitized, 144,384 were free-living parasitized, and 374,379 were parasites. The average number of organisms per sample was 75.41.

The five principal quantitative variables derived from the data were: number of samples, total sample size (number of organisms), average body mass, variance of population density, and mean of population density. The last four had highly right-skewed frequency distributions, hence these four variables were $\log_{10}$ -transformed for further analysis. Scatterplot matrices displayed the bivariate relationships of each of these five variables as a function of each of the remaining four variables, for all species combined (SI Appendix, Figure S1), and separately for free-living unparasitized species (SI Appendix, Figure S2), free-living parasitized species (SI Appendix, Figure S3), and parasitic species (SI Appendix, Figure S4). These summary graphics provided a useful check for erroneous outliers and a visual impression of which variables were most closely associated.

For each lifestyle, by visual inspection the linear relationship between log variance and log mean density was much tighter than the linear relationship of any other pair of the five principal

quantitative variables (SI Appendix, Figures S2-S4). Log mean density clearly increased with log total sample size (log total number of organisms), as would be expected since log mean density is the total number of organisms per unit of area. Since log variance was very nearly linearly associated with log mean density, it was not surprising that log variance also increased with log total sample size.

Statistical calculations used JMP version 10 (1) and MATLAB version 8.3.0.532 R2014a (2). Analysis of covariance was used to test for differences of slopes. Analysis of variance was used to test for significance of effects. Quantile regression was used to estimate DMA and VMA. Quantile linear regression estimates the parameters of a straight line that lies above a certain percentage, in our case 90%, of the y-values conditional on each value of the predictor x-value. The MATLAB function `quantreg.m` (3) estimated each parameter of DMA and VMA with confidence intervals based on bootstrap sampling. As a check, the MATLAB function `rq_fnm.m` (<http://www.econ.uiuc.edu/~roger/research/rq/rq.m>, accessed 2014-09-12) was also run on the same data. The two programs gave estimates of the slope that generally agreed exactly to four decimal places but occasionally differed by 0.0001. They gave estimates of the intercept that generally agreed exactly to four decimal places but occasionally differed by 0.0001 or 0.0002. These differences in estimates are immaterial. Because `rq_fnm.m` gave no confidence intervals for the parameter estimates, we reported below only the estimates of parameters and confidence intervals from `quantreg.m`.

To test whether a putative linear relationship provided a statistically satisfactory description of a set of (x, y) observations, a quadratic relationship was fitted by ordinary least squares. If the coefficient of the quadratic term was statistically significantly different from zero ($P < 0.05$), then the nonlinearity was considered significant at the 0.05 level. For example, the quadratic extension of TL was

$$\log \text{ variance} = \log(a) + b \times \log \text{ mean} + c \times (\log \text{ mean}) \times (\log \text{ mean})$$

which is mathematically identical to (though differently expressed than) equation (14) of Taylor, Woiwod, and Perry (4). Here and in all calculations, $\log = \log_{10}$.

Possibilities for future research

We measured the population density of parasitic species as individuals per square meter, as we did for all free-living species, and not as individuals per host, as in some prior studies of TL in parasitology (e.g., 5). Did the choice of measure materially affect the conclusions drawn? Do both measures satisfy TL? If so, is there a connection (theoretical or empirical) between the TL exponent using parasitic individuals per square meter and the TL exponent (if TL applies) using parasitic individuals per host individual?

Preliminary analyses not reported here indicated that the frequency distribution of parasites per host would be well described by the negative binomial distribution in many instances. But for

many parasites of fish, the prevalence of infection is 100% of hosts so the frequency distribution of parasites per host includes no zero counts, contrary to the negative binomial distribution. Relating the underlying distribution of parasites per host to the form and parameters of TL, if TL is satisfied, remains a challenge.

Mathematically, any two of the three power laws TL, DMA, and VMA predict the third, but statistically not all pairs of these power laws are equally useful in predicting the remaining power law. Specifically, in DMA (Figure 2) and VMA (Figure 3), for each value of body mass, the mean and variance, respectively, of population density range widely (by multiple orders of magnitude). Without additional information to connect particular values of mean density with particular values of the variance in density for a given value of body mass, it would not be possible to reconstruct the relatively tight association between the mean and the variance of population density in TL (Figure 1). However, given the similarities in the (rescaled) scatter of log mean and log variance of population density for a given average body mass (SI Appendix, Figure S6), it is not obvious whether the mechanisms underlying TL and DMA generate VMA or the mechanisms underlying TL and VMA generate DMA. This remains a question for future theoretical and empirical research.

Supplementary references

1. SAS Institute Inc. (2012) *Using JMP 10* (SAS Institute Inc., Cary, NC).
2. Mathworks (2014) *MATLAB R2014a* (Mathworks Inc., Natick, MA).
3. Grinstead A (2011) quantreg.m: Quantile regression with bootstrapping confidence intervals. Updated 08 July 2011. <http://www.mathworks.com/matlabcentral/fileexchange/32115-quantreg-m-quantile-regression>, accessed 2014-09-11.
4. Taylor LR, Woiwod IP, Perry JN (1978) The density-dependence of spatial behaviour and the rarity of randomness. *J Anim Ecol* 47:383-406. (Stable URL: <http://www.jstor.org/stable/3790>)
5. Morand S, Guégan JF (2000) Distribution and abundance of parasite nematodes: ecological specialisation, phylogenetic constraint or simply epidemiology? *Oikos* 88:563–573. (doi:10.1034/j.1600-0706.2000.880313.x)

Supplementary figures S1-S6

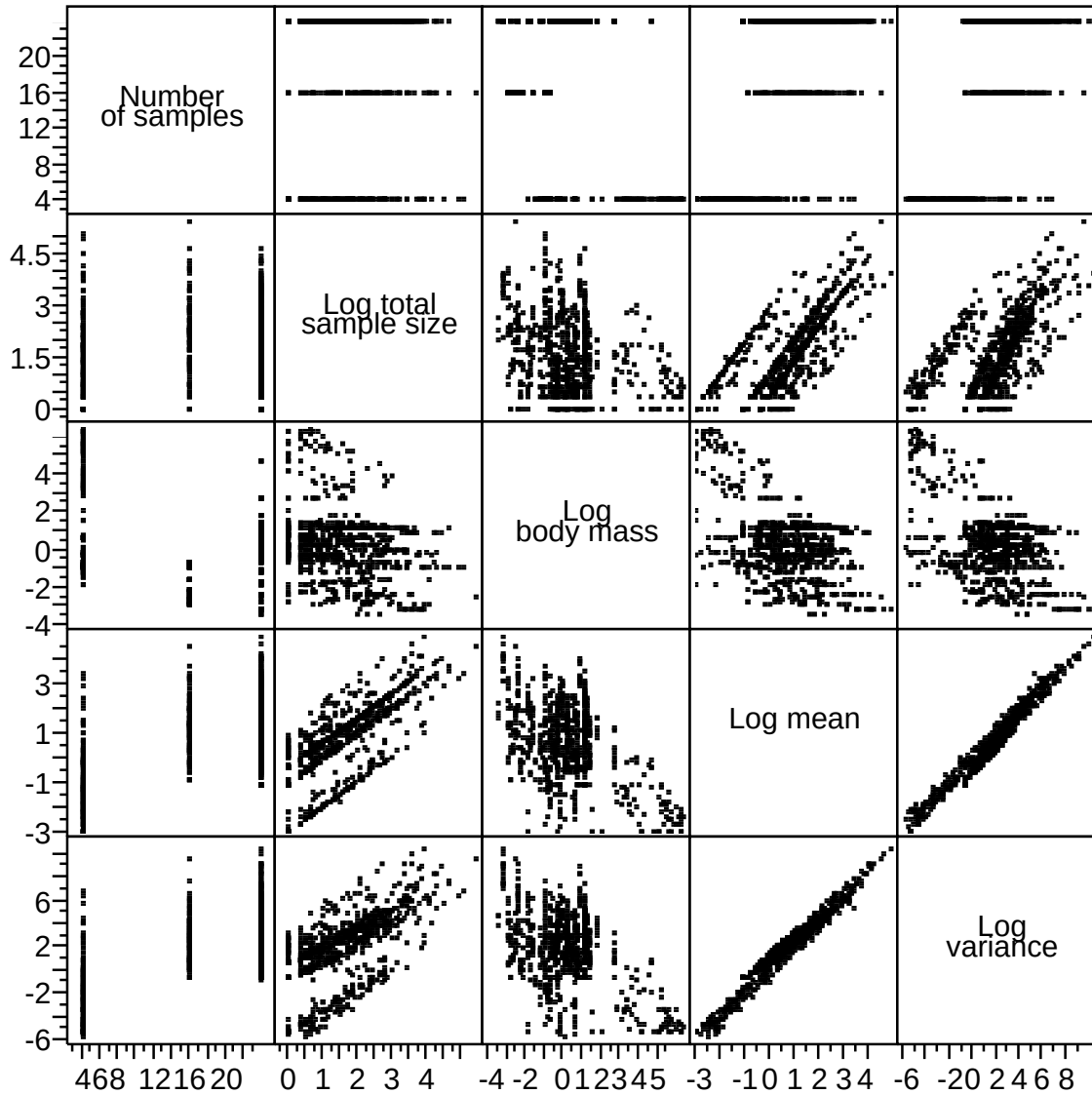


Figure S1. Scatterplot matrix of number of samples, log total sample size, log body mass, log mean of population density, and log variance of population density as a function of each of the other four variables, for species of all lifestyles pooled. For example, the scatterplot in the fifth row from the top and fourth column from the left displays log variance as a function of log mean, which is a test of Taylor's law. The scatterplot in the fourth row from the top and third column from the left displays log mean as a function of log body mass, which is a test of density-mass allometry. Each scatterplot has 733 points.

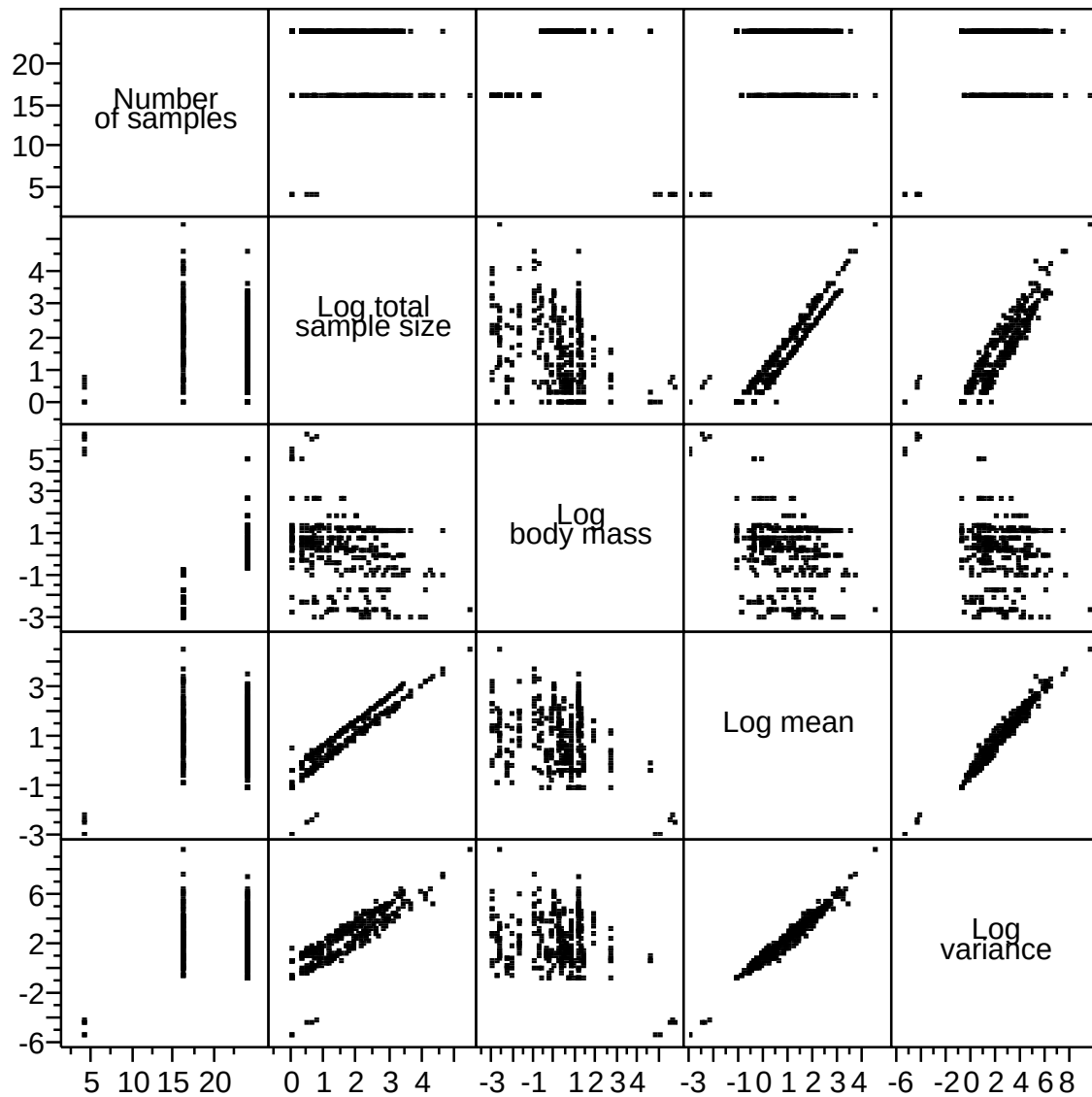


Figure S2. Scatterplot matrix of number of samples, log total sample size, log body mass, log mean of population density, and log variance of population density as a function of each of the other four variables, for unparasitized free-living species. The scatterplots are laid out as in Figure S1. Each scatterplot has 329 points.

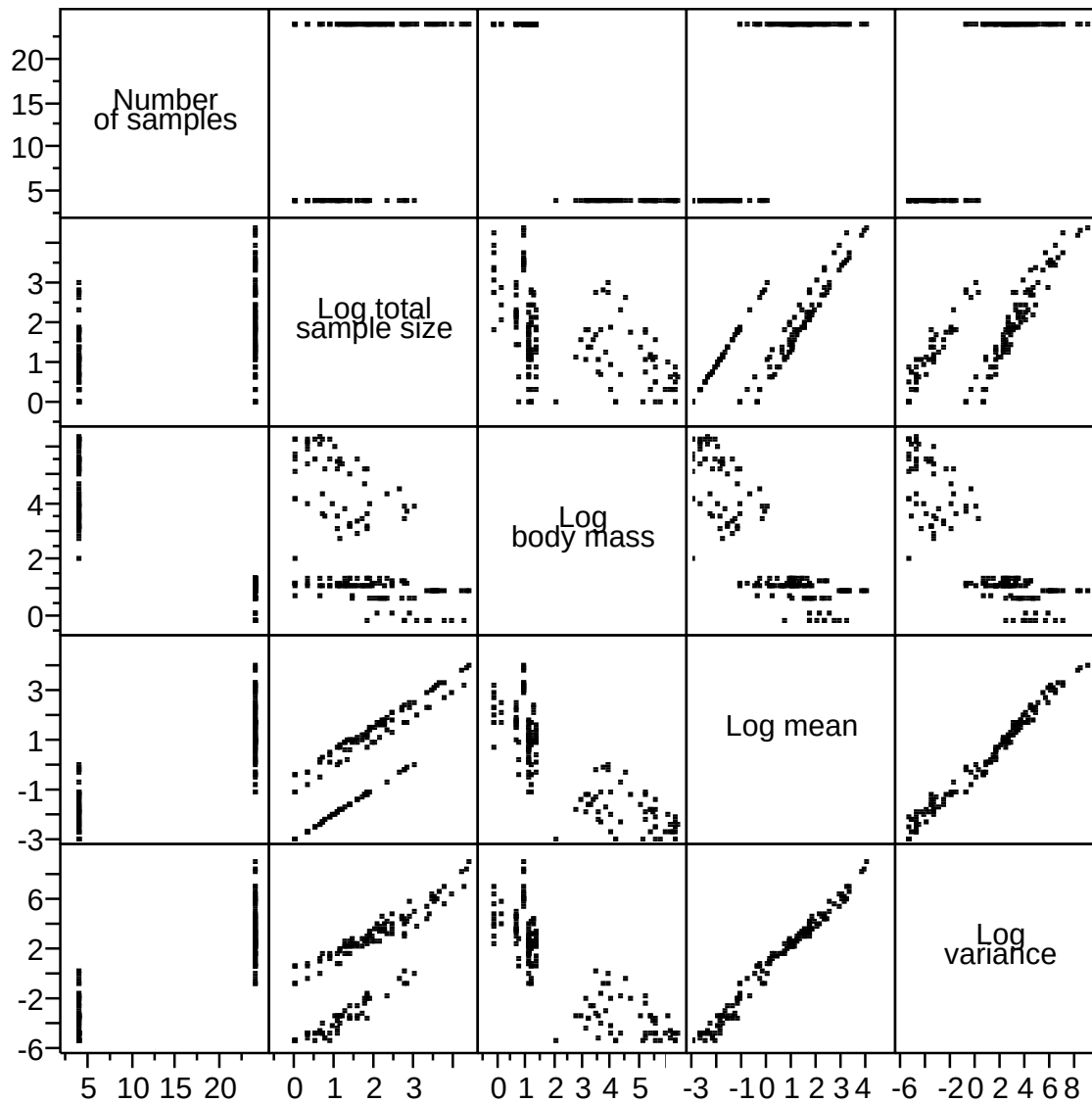


Figure S3. Scatterplot matrix of number of samples, log total sample size, log body mass, log mean of population density, and log variance of population density as a function of each of the other four variables, for parasitized free-living species. The scatterplots are laid out as in Figure S1. Each scatterplot has 151 points.

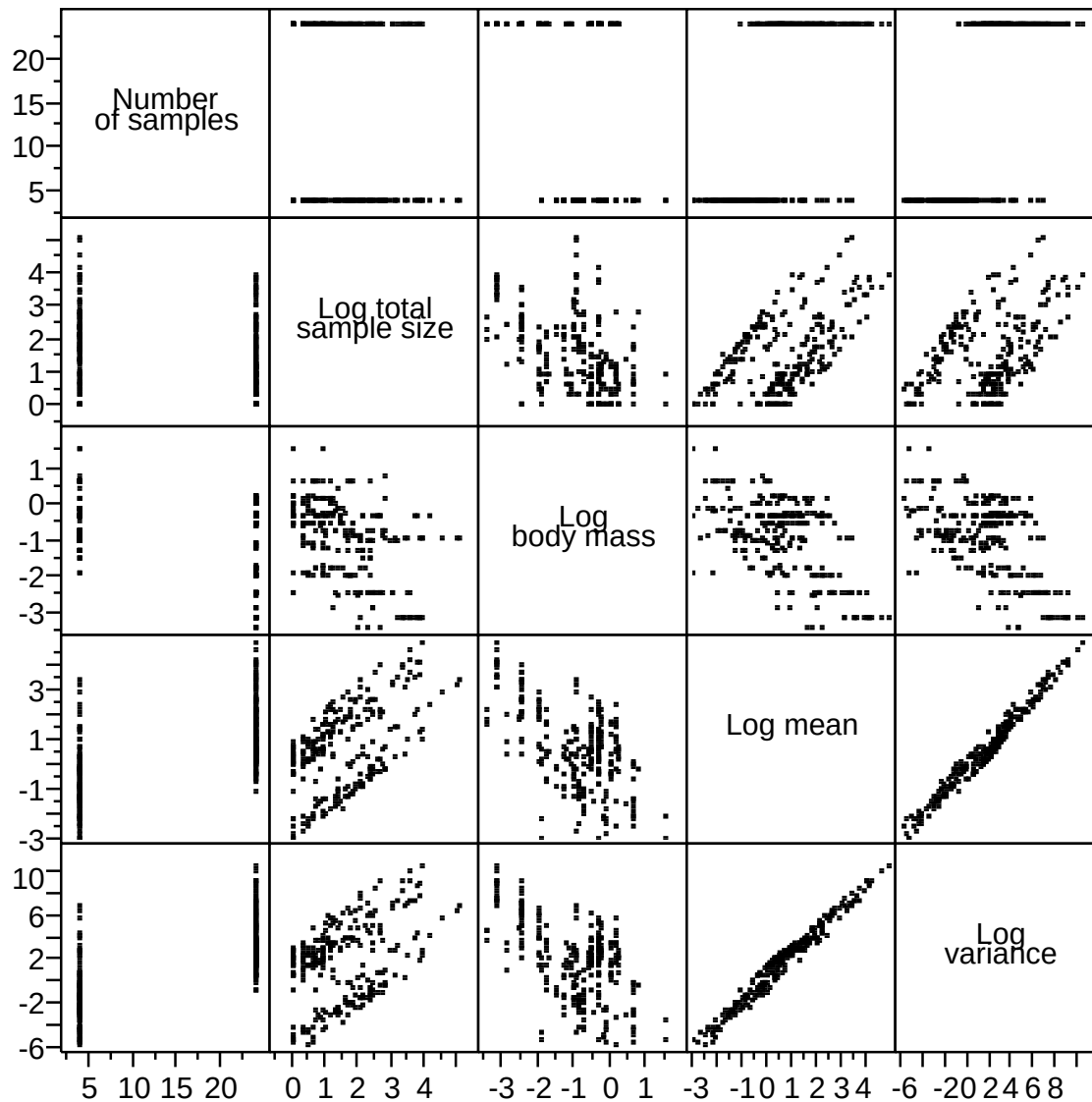


Figure S4. Scatterplot matrix of number of samples, log total sample size, log body mass, log mean of population density, and log variance of population density as a function of each of the other four variables, for parasitic species. The scatterplots are laid out as in Figure S1. Each scatterplot has 253 points.

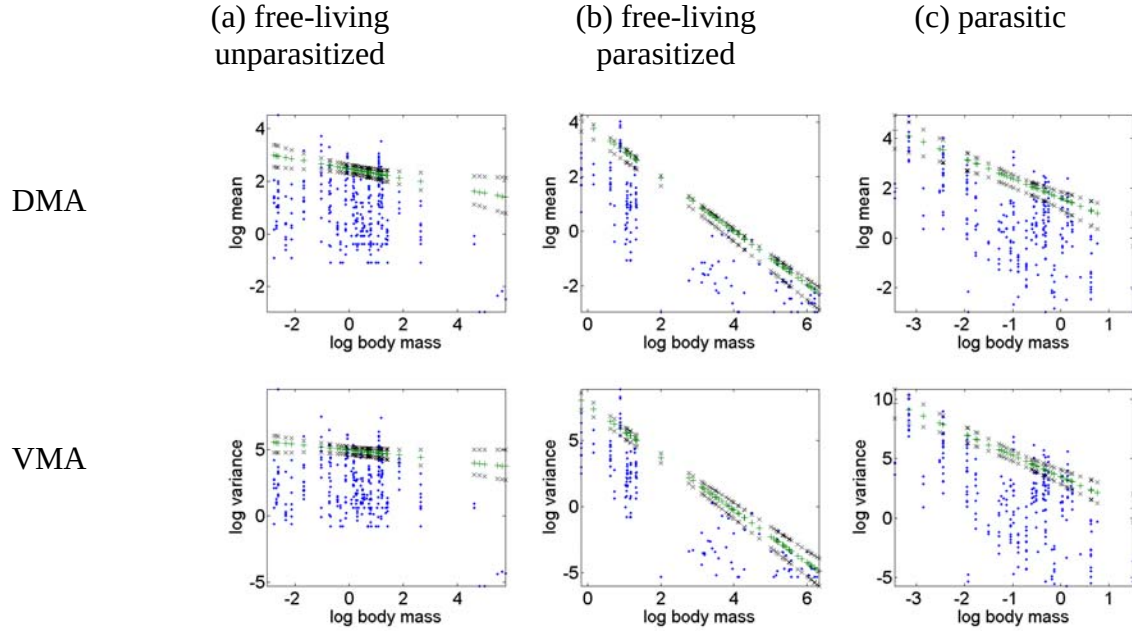


Figure S5. Quantile-regression estimation of density-mass allometry (DMA) and variance-mass allometry (VMA), for (a) free-living unparasitized species, (b) free-living parasitized species, and (c) parasites. The quantile-regression line (+ markers) is chosen to lie above 90% of the data points (small blue diamonds). The regression line is bounded above and below by its 95% confidence limits (x markers). The estimated parameters and their 95% confidence intervals are given in Table S1.

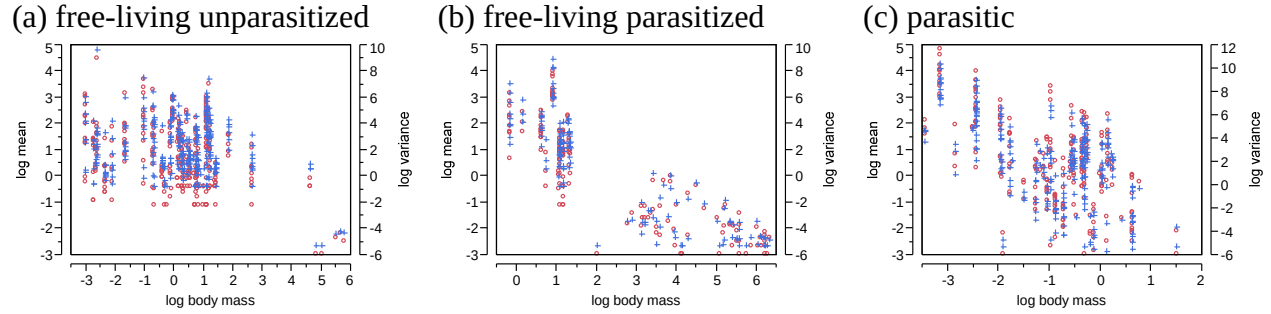


Figure S6. Log mean population density (left scales, open red circles $\circ$) and log variance of population density (right scales, blue plus signs $+$) as functions of log mean body mass, for (a) free-living unparasitized species, (b) free-living parasitized species, and (c) parasites, using independent vertical scales for the log mean (left, $\circ$) and log variance (right, $+$). The generally close pairing of data markers for the log mean (left, $\circ$) and log variance (right, $+$) reflects TL, and the general downward trend of both markers with increasing log body mass reflects DMA and VMA. In panels (a) and (b), the left scale runs from -3 to +5, while the right scale runs from -6 to +10, which is just 2 times the values in the left scale, because the slope of TL is not far from 2. In panel (c), the vertical scale on the right covers -6 to +12, which is a bit more than twice the range of the vertical scale on the left from -3 to +5, because the slope of TL slightly exceeds 2 for parasites.

Table S1. Parameter estimates using quantile regression to fit density-mass allometry (DMA) and variance-mass allometry (VMA) to log-transformed data on the spatial variation of population density (individuals per square meter) of free-living unparasitized species, free-living parasitized species, and parasitic species. The quantile-regression line is chosen to lie above 90% of the data points. For all three lifestyles, the intercept predicted for VMA, $\log_{10}(a)+b\times\log_{10}(u)$, fell within the 95% confidence interval of the intercept estimated for VMA, and the slope predicted for VMA, $b\times v$, fell within the 95% confidence interval of the slope estimated for VMA. Abbreviations: lo, Lower limit of 95% confidence interval, hi, Upper limit of 95% confidence interval

Lifestyle→	Free-living unparasitized	Free-living parasitized	Parasitic
Parameter ↓			
n : number of samples	329	151	253
DMA: mean density = $u\times(\text{mean body mass})^v$			
$\log_{10} u$	2.4784	3.9095	1.5903
Lo	2.3146	3.6055	1.2565
Hi	2.6422	4.2135	1.9242
v	-0.1849	-0.9784	-0.7903
Lo	-0.2995	-1.0744	-1.024
Hi	-0.0702	-0.8824	-0.5567
VMA: variance = $c\times(\text{mean body mass})^d$			
$\log_{10} c$	4.9852	7.6867	3.4812
Lo	4.6848	7.1332	2.91
Hi	5.2856	8.2403	4.0523
$\log_{10}(a)+b\times\log_{10}(u)$	4.9702	8.1848	3.7761
d	-0.2127	-1.9878	-1.788
Lo	-0.4109	-2.1769	-2.2269
Hi	-0.0144	-1.7987	-1.3491
$b\times v$	-0.3107	-1.9757	-1.6612